SMART INDIA HACKATHON 202

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TITLE PAGE

Problem Statement ID – 26167

Problem Statement Title- SatQuery AI – An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

Theme- Space Technology

PS Category- Software

Team ID- ‹fill from portal›

Team Name (Registered on portal) - TMS<3

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IDEA TITLE | SatQuery AI

Proposed Solution (Describe your Idea/Solution/Prototype)

Ask a plain-English question about any satellite scene. An agent validates the inputs, picks the right remote-sensing specialist models, runs them, and answers with visual evidence — no GIS skill, no model selection, no parameters.

NL Query + Image(s) — GeoTIFF · SAR · optical–SAR · bi-temporal → **Input Validator** — modality · CRS · GSD · co-registration → **Agentic Task Router** — classify query → plan tool chain → **RS Specialist Registry** — VQA · Ground · Change · Optical-SAR → **Evidence Fusion** — merge text + masks, score confidence → **Grounded Answer** — overlay + report + audit trail

Detailed explanation of the proposed solution

- Web GUI accepts **one image, a co-registered optical–SAR pair, or a bi-temporal pair** plus a free-text question.
- A router (small LLM + rule guard) classifies the query into VQA / caption / grounding / change / cross-modal.
- Four **remote-sensing-adapted specialists** are called in sequence — not one generic VLM answering everything.
- LoRA adapters trained on **BigEarthNet.txt** (464K co-registered S1+S2 scenes, 9.6M text annotations).
- Output = answer text + mask/box overlay + confidence + one-click PDF & GeoJSON report.

How it addresses the problem

- **Non-experts get answers:** no band maths, model choice or GIS workflow — just a question.
- **Answers a single optical image cannot:** SAR sees through cloud and at night; bi-temporal pairs reveal change.
- **One system, five tasks** replaces today's isolated, single-task remote-sensing tools.
- **Domain-adapted, not generic:** sensor terms, polarisation and land-cover vocabulary are learned, not guessed.
- **Auditable** execution summary (task, models, parameters) makes it usable in operational ISRO/SAC workflows.

Innovation and uniqueness of the solution

- **Query-driven agentic orchestration** over a tool registry — the question decides the pipeline, not the developer.
- **Cross-modal optical + SAR reasoning** inside a single conversational interface.
- **Confidence gating with abstention:** it refuses and explains on mis-registered or wrong-modality inputs instead of hallucinating.
- **Evidence-first design** — every claim is tied to a mask, box or band statistic shown to the user.
- **Adapter-swap architecture:** four specialists, one backbone — the whole stack fits on a single 16 GB GPU.

Representative query → workflow the agent selects automatically

"Describe the land-cover and major objects in this image." ▸ **Captioning + RS-VQA**	"Highlight the water body referred to in the query." ▸ **Text-guided grounding → box / mask**	"What changed between these two dates, and where?" ▸ **Change VQA + change map**	"Use optical and SAR together to identify built-up and water." ▸ **Optical–SAR fusion extraction**	"Has the built-up area increased, decreased or stayed the same?" ▸ **Change VQA + area statistics**

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TECHNICAL APPROACH

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Technologies to be used (e.g. programming languages, frameworks, hardware)

LANGUAGES Python 3.11 · TypeScript · CUDA

VLM BACKBONE Qwen2-VL / TinyLLaVA (2–7B) + RemoteCLIP & GeoChat-class RS encoders

ADAPTATION LoRA / QLoRA via PEFT · HuggingFace Transformers · PyTorch · DeepSpeed

SPECIALIST TOOLS ChangeFormer / BiT change nets · Grounding-DINO + SAM masks · dual-encoder optical–SAR fusion

AGENT LAYER LangGraph controller · JSON-schema tool registry · FastAPI · Celery + Redis

GEOSPATIAL GDAL / rasterio · pyproj · GeoPandas · SNAP speckle filter · Leaflet + deck.gl

FRONTEND React + Vite + Tailwind · split-screen image viewer · trace panel

HARDWARE Train: Colab / Kaggle T4–A100 (≈ 25 GPU-h per adapter) · Serve: 1× 16 GB GPU, 4-bit; ONNX CPU fallback · Docker

FIVE TASK FAMILIES THE ROUTER CHOOSES FROM

- Single-image VQA (mandatory baseline)
- Captioning / scene description
- Text-guided region grounding
- Bi-temporal change VQA + change map
- Optical–SAR joint information extraction

AUDIT TRACE (auto-generated per query)

{ task: change_vqa · tools: [validator, changeformer, change_vqa_lora] · params: {thr: 0.5, gsd: 10m} · confidence: 0.87 }

Methodology and process for implementation (Flow Charts/Images/ working prototype)

1 · INPUT & VALIDATION

- **Upload** — GeoTIFF / TIFF · PNG-JPEG for benchmarks
- **Compatibility check** — CRS · GSD · bands · extent · modality
- **Pair check** — co-registration + acquisition dates

2 · AGENTIC CONTROLLER

- **Query understanding** — intent + entities
- **Task classifier** — 5 task families
- **Tool planner** — select + sequence
- **Parameter binder** — allow-listed only

3 · RS SPECIALIST REGISTRY

- **RS-VQA** — single-image · mandatory
- **Caption + Grounding** — text-guided region boxes
- **Change VQA** — bi-temporal + change map
- **Optical–SAR Fusion** — complementary extraction

4 · FUSION & OUTPUT

- **Cross-check + confidence** — agreement across tools, abstain if low
- **Visual evidence** — mask / box / change-map overlay
- **Execution trace** — task, models, params → PDF + GeoJSON

ADAPTATION & EVALUATION

BigEarthNet.txt (464K S1+S2 · 9.6M texts · CC-BY 4.0) → **LoRA / QLoRA** (< 1% params trained) → **Task heads** (VQA · ground · change · fusion) → **Benchmark eval** (VRSBench · RSVQA · CDVQA) → **Versioned registry** (model card + metrics)

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

✓ **Data is free and ready.** BigEarthNet.txt (CC-BY 4.0), VRSBench, RSVQA and CDVQA are all open — zero acquisition cost or licensing delay.

✓ **No model built from scratch.** Open-weight RS encoders + a small VLM backbone; we train only LoRA adapters (< 1% of parameters).

✓ **Fits our compute.** ≈ 25 GPU-hours per adapter on free T4 / A100 sessions; 4-bit inference on a single 16 GB GPU.

✓ **Low research risk.** Each of the four capabilities already has published baselines and public leaderboards to match.

✓ **Orchestration is plain software.** FastAPI + LangGraph + a JSON tool registry — deterministic, testable, no novel research.

✓ **Demo-ready in 12 weeks.** Vertical slice (upload → VQA → overlay) working by week 5, full agent by week 10.

VIABILITY BEYOND THE HACKATHON

- Fully open-source stack — deployable on-premise at SAC, no vendor lock-in.
- Adapters retrain per sensor: a new ISRO mission plugs in without rebuilding the agent.
- Model cards + versioned registry keep every answer reproducible.
- Containerised REST API — same backend can serve Bhuvan / Bhoonidhi.

Potential challenges and risks → Strategies for overcoming these challenges

Potential challenges and risks	Strategies for overcoming these challenges
SAR is not optical speckle, geometry, far fewer SAR–text pairs than RGB	Train on BigEarthNet.txt's co-registered S1+S2 pairs; speckle filtering + dB scaling; separate SAR encoder branch with modality prompt tokens
Compute ceiling a full fine-tune of a large VLM is out of reach	LoRA/QLoRA on a 2–7B backbone, gradient checkpointing, 4-bit serving; one shared backbone with four swappable adapters
Hallucinated answers a VLM can produce fluent but ungrounded text	Confidence gating with an explicit abstain path; every claim bound to a mask, box or band statistic; trace shown to the user
ISRO/SAC domain gap Cartosat-2S + RISAT differ from Sentinel in GSD and radiometry	Sensor-agnostic preprocessing (radiometric normalisation, resample to fixed GSD), a thin band-adapter layer, test-time augmentation
Bad or mismatched uploads wrong modality, mis-registration, missing metadata	Hard input validator with auto co-registration; the system refuses and explains rather than answering on invalid input

12-WEEK EXECUTION ROADMAP

1. **W1–2 · Data & baselines** — loaders, splits, metric harness
2. **W3–5 · RS adaptation** — LoRA on BigEarthNet.txt → VQA + caption/grounding
3. **W6–8 · Pair reasoning** — change VQA + change map, optical–SAR fusion
4. **W9–10 · Agent + GUI** — router, registry, overlays, audit trace
5. **W11–12 · Evaluate & harden** — VRSBench/RSVQA/CDVQA, latency, Docker

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IMPACT AND BENEFITS

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Potential impact on the target audience

SatQuery AI — one natural-language interface over five remote-sensing workflows

ISRO / SAC analysts
Triage a Cartosat–RISAT scene by asking, not by scripting a GIS chain.

State disaster authorities
Flood and cyclone extent through cloud cover, day or night, via SAR.

Agriculture departments
Crop-cover and irrigation questions answered per district, per season.

Urban local bodies
"Has built-up area grown here since 2022?" with a change map as proof.

Forest & water boards
Deforestation and water-body shrinkage tracked from bi-temporal pairs.

Students & researchers
Entry-level access to EO analysis with no GIS or ML background.

Benefits of the solution (social, economic, environmental, etc.)

SOCIAL

- Disaster answers in minutes, not an analyst-day
- EO data usable by officials with no GIS training
- Query layer is language-agnostic — extendable to Indian languages
- Same tool serves a district officer and a PhD researcher
- Visual evidence makes decisions explainable to the public

ECONOMIC

- One deployment replaces 4+ siloed single-task tools
- Cuts specialist analyst hours per query request
- Open weights + commodity 16 GB GPU — no per-seat licence
- Faster insurance / relief assessment after a disaster event
- Zero data-acquisition cost — Sentinel + ISRO archives are open

ENVIRONMENTAL

- Routine deforestation and encroachment monitoring at scale
- Water-body and glacier-extent change tracked over time
- Urban-sprawl quantification to support planning policy
- Crop-stress and irrigation checks without a field visit
- Low-power inference — one GPU, not a training cluster

STRATEGIC / NATIONAL

- Cloud-independent, day-night monitoring using indigenous RISAT SAR
- Auditable execution trace — fit for operational ISRO use
- Indigenous EO vision-language stack, no foreign API dependency
- Reusable agent pattern for any future ISRO sensor or mission
- Builds Indian capacity in EO foundation-model adaptation

TARGET OUTCOMES | match or beat published baselines on VRSBench · RSVQA · CDVQA after adaptation | < 30 s median end-to-end query latency | 100% of answers returned with visual evidence + execution trace | graceful abstention instead of a wrong answer

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

DATASETS — training & evaluation

BigEarthNet.txt — 464,044 co-registered Sentinel-1 SAR + Sentinel-2 MSI images with 9.6M text annotations (descriptions, VQA pairs, referring expressions). CC-BY 4.0. Primary adaptation corpus.
arxiv.org/abs/2603.29630 · txt.bigearth.net

VRSBench — Versatile RS benchmark — detailed captioning, visual grounding and VQA on one image set (NeurIPS 2024).
arxiv.org/abs/2406.12384 · github.com/lx709/VRSBench

RSVQA — Remote-sensing visual question answering benchmark (low/high resolution splits).
rsvqa.sylvainlobry.com

CDVQA — Change Detection Meets Visual Question Answering — bi-temporal change VQA (IEEE TGRS 2022).
github.com/YZHJessica/CDVQA

BigEarthNet-MM / reBen — Multimodal S1+S2 multi-label archive underlying BigEarthNet.txt.
bigearth.net

LEVIR-CD · SECOND — Bi-temporal change-detection datasets with reference masks for the change-map head.
chenhao.in/LEVIR · captain-whu.github.io

DOTA · DIOR-RSVG — Oriented object detection and referring-expression grounding in aerial imagery.
captain-whu.github.io/DOTA

SEN1-2 · SpaceNet-6 — Additional co-registered optical–SAR pairs for cross-modal pre-training.
spacenet.ai

MODELS & METHODS

RemoteCLIP — Vision-language foundation model for remote sensing (IEEE TGRS) — RS image–text encoder we adapt.
github.com/ChenDelong1999/RemoteCLIP

GeoChat — Grounded large vision-language model for remote sensing (CVPR 2024) — grounding + region VQA reference.
mbzuai-oryx.github.io/GeoChat

LHRS-Bot — VGI-enhanced multimodal LLM for RS imagery (ECCV 2024).
github.com/NJU-LHRS/LHRS-Bot

SkySense — Multi-modal RS foundation model for universal EO interpretation (CVPR 2024) — optical+SAR fusion design.
github.com/Jack-bo1220/SkySense

CDQAG — Show Me What and Where has Changed? — QA + grounding for change detection.
arxiv.org/abs/2410.23828

LoRA / QLoRA · SAM — Parameter-efficient adaptation and promptable mask refinement.
PEFT · segment-anything

ChangeFormer · BiT — Transformer change-detection backbones producing the spatial change map.
github.com/wgcban/ChangeFormer

Grounding DINO — Open-set text-conditioned detection used for text-guided region grounding.
github.com/IDEA-Research/GroundingDINO

Qwen2-VL · TinyLLaVA — Compact open VLM backbones that fit our GPU budget after 4-bit quantisation.
huggingface.co

Change-Agent — Agentic change interpretation over bi-temporal RS imagery — closest prior art.
github.com/Chen-Yang-Liu/Change-Agent

SURVEYS, TOOLING & ISRO CONTEXT

VLM meets Remote Sensing — Survey of models, datasets and open problems in RS vision-language modelling.
arxiv.org/abs/2505.14361

Awesome RS Foundation Models — Curated index of open RS foundation-model weights and benchmarks.
github.com/Jack-bo1220/Awesome-Remote-Sensing-Foundation-Models

Cartosat-2S · RISAT-1/2B — Indian optical and SAR product specifications and sample data for the ISRO/SAC evaluation domain.
bhoonidhi.nrsc.gov.in · nrsc.gov.in

Copernicus Open Data — Sentinel-1 / Sentinel-2 archive used for additional pre-training tiles.
dataspace.copernicus.eu

Engineering stack — GDAL / rasterio, ESA SNAP (speckle filtering), LangGraph, FastAPI, PyTorch + PEFT.
official project docs

RS-MLLM survey index — Running index of multimodal LLMs for remote sensing and their benchmarks.
github.com/ZhanYang-nwpu/Awesome-RS-MLLM

Bhuvan · Bhoonidhi — ISRO geoportals — target integration surface for the deployed service.
bhuvan.nrsc.gov.in

Evaluation protocol — Prescribed public test splits, normalised metric combination as stated in the PS.
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